

Chemistry marking guide and response

External assessment 2025

Combination response (110 marks)

Assessment objectives

This assessment instrument is used to determine student achievement in the following objectives:

1. describe and explain chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design
2. apply understanding of chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design
3. analyse evidence about chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design to identify trends, patterns, relationships, limitations or uncertainty
4. interpret evidence about chemical equilibrium systems, oxidation and reduction, properties and structure of organic materials, and chemical synthesis and design to draw conclusions based on analysis.

Note: Objectives 5, 6 and 7 are not assessed in this instrument.

Purpose

This document consists of a marking guide and a sample response.

The marking guide:

- provides a tool for calibrating external assessment markers to ensure reliability of results
- indicates the correlation, for each question, between mark allocation and qualities at each level of the mark range
- informs schools and students about how marks are matched to qualities in student responses.

The sample response demonstrates the qualities of a high-level response.

Mark allocation

Where a response does not meet any of the descriptors for a question or a criterion, a mark of '0' will be recorded.

Allow FT mark/s — refers to 'follow through', where an error in the prior section of working is used later in the response, a mark (or marks) for the rest of the response can still be awarded so long as it still demonstrates the correct conceptual understanding or skill in the rest of the response.

Marking guide

Multiple choice

Question	Response
1	A
2	C
3	D
4	C
5	C
6	D
7	C
8	C
9	D
10	B
11	A
12	B
13	B
14	D
15	C
16	B
17	B
18	A
19	A
20	D

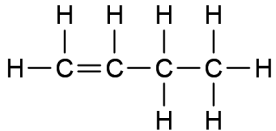
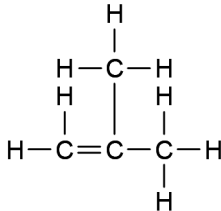
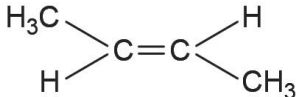
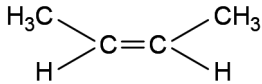
Short response: Paper 1

Q	Sample response	The response:	Notes
21a)	HA is a weak acid.	classifies HA as a weak acid [1 mark]	
21b)	HA is a monoprotic acid because it donates one hydrogen ion.	explains that HA is a monoprotic acid because it donates one hydrogen ion [1 mark]	For <i>hydrogen ion</i> , also accept proton or H ⁺ .

Q	Sample response	The response:	Notes
22	Oxidation state (Mn): +3 Name of ionic compound (Mn_2O_3): manganese(III) oxide	- determines oxidation state is +3 [1 mark] - determines ionic name is manganese(III) oxide [1 mark]	For <i>oxidation state</i> , do not accept 3+. Allow FT mark for name correctly based on incorrect oxidation state.

Q	Sample response	The response:	Notes
23	1: All solutions are 1.0 mol L ⁻¹ . 2: Temperature is 25 °C (298 K).	- identifies one limitation [1 mark] - identifies a second limitation [1 mark]	For <i>limitation</i> , also accept: - gases must be at a pressure of 100 kPa - substances must be pure - platinum is used, as the electrode if the half-cell does not include a solid metal.

Q	Sample response	The response:	Notes
24a)	Unsaturated because it contains a double bond	- classifies citronellol as unsaturated [1 mark] - explains that it contains a double bond [1 mark]	
24b)	Bromine water could be used to confirm that citronellol is unsaturated. When added to citronellol, bromine water will change from brown/orange to colourless because the molecule is unsaturated.	- identifies a suitable chemical test [1 mark] - describes a suitable chemical test [1 mark]	For a <i>suitable chemical test</i> also accept - potassium permanganate solution - or other suitable response consistent with a reasonable understanding. For <i>describes a suitable chemical test</i> - also accept - the change of the permanganate solution from deep purple to a brown precipitate - the colour change of the acidified permanganate solution from deep purple to colourless. - do not accept - the bromine water becomes clear.

Q	Sample response	The response:	Notes
25a)	 IUPAC name: but-1-ene	- draws the structural formula of a structural isomer [1 mark] - identifies IUPAC name of the structural isomer [1 mark]	For <i>structural isomer</i>, also accept:  IUPAC name: 2-methylpropene For <i>structural isomer</i>, also accept condensed structural formulas: - but-1-ene: $\text{CH}_2\text{CHCH}_2\text{CH}_3$ 2-methylpropene: $\text{CH}_2\text{C}(\text{CH}_3)_2$. Allow FT mark for correct IUPAC name based on incorrect structural formula.
25b)	 IUPAC name: (E)-but-2-ene	- draws the structural formula of a geometric isomer [1 mark] - identifies IUPAC name of the geometric isomer [1 mark]	For <i>geometric isomer</i>, also accept:  IUPAC name: (Z)-but-2-ene For (E)-but-2-ene also accept <i>trans</i>-but-2-ene. For (Z)-but-2-ene also accept <i>cis</i>-but-2-ene. Allow FT mark for correct IUPAC name based on incorrect structural formula.

Q	Sample response	The response:	Notes
26a)	The forward reaction is exothermic.	determines that forward reaction is exothermic [1 mark]	
26b)	Increasing pressure would shift equilibrium to the products to produce more SO ₃ (g). The reaction would shift towards the products to reduce pressure because there are fewer gas molecules as products (2 molecules) than reactants (3 molecules).	- determines that increasing pressure increases production of SO₃(g) [1 mark] - explains that equilibrium shifts to the side with fewer molecules to reduce pressure [1 mark]	

Q	Sample response			The response:	Notes
27					
	Electrolyte	Cathode product	Oxidation half-equation		
	concentrated (25%) NaCl (aq)	H ₂ (g)	2Cl ⁻ (aq) ⇌ Cl ₂ (g) + 2e ⁻	- determines H₂(g) as product at cathode for concentrated (25%) NaCl(aq) [1 mark] - determines oxidation half-equation for concentrated (25%) NaCl(aq) [1 mark]	
	molten NaCl(l)	Na(l)	2Cl ⁻ (l) ⇌ Cl ₂ (g) + 2e ⁻	- determines Na(l) as product at cathode for molten NaCl(l) [1 mark] - determines oxidation half-equation for molten NaCl(l) [1 mark]	

Q	Sample response	The response:	Notes
28a)	$pK_w = pK_a + pK_b$ $pK_b = 14 - 9 = 5$	determines that pK_b equals 5 [1 mark]	Accept 4.9 to 5.1 inclusive.
28b)	Moles of BOH in 25.0 mL aliquot of diluted base = moles HCl at equivalence point	determines moles of base in aliquot equals moles of HCl at equivalence point [1 mark]	
	moles of BOH = $0.1 \times 0.020 = 2.0 \times 10^{-3}$ moles $[\text{diluted BOH}] = \frac{2.0 \times 10^{-3}}{0.025} = 0.08 \text{ mol L}^{-1}$	calculates concentration of BOH in aliquot [1 mark]	Also accept: $C_{\text{HCl}} V_{\text{HCl}} = C_{\text{BOH}} V_{\text{BOH}}$ $C_{\text{BOH}} = \frac{0.1 \times 0.020}{0.025} = 0.08 \text{ mol L}^{-1}$
	$[\text{BOH}] = 0.08 \times 5$ $= 0.40 \text{ mol L}^{-1}$	calculates concentration of BOH [1 mark]	Allow FT mark for correct calculation of [BOH] based on incorrect [diluted BOH].
28c)	The pH of the equivalence point: would increase to pH 7 because NaOH is a strong base. The volume of HCl required to reach the equivalence point: would remain the same, as the moles of base being neutralised has not changed.	- determines that the pH at the equivalence point is 7 [1 mark] - explains that the pH at the equivalence point would increase when a strong base is titrated with a strong acid [1 mark] - determines that the volume of HCl required to reach the equivalence point would not change [1 mark] - explains that the moles of base to be neutralised remains the same [1 mark]	

Q	Sample response	The response:	Notes
29a)	Another principle of green chemistry used in the production of benzaldehyde is prevention, because it is better to prevent waste than to treat or clean it up after it has been created.	- identifies prevention as a green chemistry principle [1 mark] - explains that it is better to prevent waste than treat or clean it up after it has been created [1 mark]	For <i>green chemistry principle</i>, also accept catalysis. For an <i>explanation</i>, also accept catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
29b)	The Na-MnO_x catalyst reaction pathway has the best atom economy, as it maximises the conversion of reactants into benzaldehyde. $AE = \frac{\text{MW of desired product}}{\text{MW of reactants}}$ $AE = \frac{212.26 \text{ g mol}^{-1}}{230.3 \text{ g mol}^{-1}} = 0.92$ Meanwhile, the Criegee oxidation reaction pathway has the lowest E-factor, indicating that it has the least environmental impact because it produces the least waste (3.2 kg) per kg of benzaldehyde produced.	- concludes that the Na-MnO_x catalyst pathway has best atom economy [1 mark] - provides evidence to support atom economy claim [1 mark] - concludes that the Criegee oxidation pathway has the lowest environmental impact [1 mark] - provides E-factor evidence to support environmental impact claim [1 mark]	For <i>the Criegee oxidation pathway has the lowest environmental impact</i>, also accept the Malaprade oxidation reaction pathway has the greatest environmental impact. For <i>E-factor evidence</i>, also accept the Criegee reaction pathway produces the largest amount of benzaldehyde per kg of waste.

Short response: Paper 2

Q	Sample response	The response:	Notes
1a)	$\text{Mg(s)} \rightleftharpoons \text{Mg}^{2+}(\text{aq}) + 2\text{e}^{-}$	determines the oxidation half-equation for the Mg $\text{Mg}(\text{NO}_3)_2$ half-cell [1 mark]	
1b)	Electrons flow from the magnesium electrode to the copper electrode via the wire, while anions move through the salt bridge towards the magnesium half-cell.	- identifies that electrons flow from the magnesium electrode to the copper electrode through the wire [1 mark] - identifies that anions move through the salt bridge towards the magnesium half-cell [1 mark]	Allow FT mark for correct electron flow based on incorrect response to Q1a). For <i>electrons flow</i>, do not accept from the anode to the cathode (identification of the metal is required). For <i>the anions move</i>, do not accept towards the anode (identification of the half-cell is required).
1c)	The copper electrode would increase in mass, and the copper nitrate solution would become lighter in colour.	- describes a change observed in the cell [1 mark] - describes a second change observed in the cell [1 mark]	For observed change accept: – mass of copper electrode would increase – mass of magnesium electrode would decrease – voltage would decrease. Allow FT mark from Q1a) for changes correctly based on incorrect identification of the electrode.
1d)	A voltage will be produced because Al is a more reactive metal than Cu. Therefore Al will be oxidised and Cu^{2+} will be reduced.	- predicts that a voltage would be produced [1 mark] - identifies that Al is more reactive than Cu [1 mark] - explains that Al would be oxidised and Cu^{2+} would be reduced [1 mark]	For <i>explanation</i>, accept: – half-reactions – standard electrode potentials (E^0) – relative position in standard electrode potentials table.

Q	Sample response	The response:	Notes
2a)	Hexanal would produce a CHO^+ fragment, while hexan-2-one would produce a CH_3CO^+ fragment.	- describes that hexanal would produce a CHO^+ fragment [1 mark] - describes that hexan-2-one would produce a CH_3CO^+ fragment [1 mark]	For CH_3CO^+ , also accept $\text{C}_2\text{H}_3\text{O}^+$. Award 1 mark maximum if fragments are missing positive charge.
2b)	IR spectrum for hexanoic acid would have an OH^- broad peak around $2500\text{--}3000\text{ cm}^{-1}$ while hexanal would not.	explains that hexanoic acid would also have a peak at $2500\text{--}3000\text{ cm}^{-1}$ for OH group [1 mark]	

Q	Sample response	The response:	Notes
3a)	Indicators are weak acids, therefore, $K_a(\text{alizarin yellow R}) = \frac{[\text{H}_3\text{O}^+][\text{In}^-]}{[\text{HIn}]} = 7.9 \times 10^{-12}$ $\text{pH} = -\log[\text{H}_3\text{O}^+] = -\log[7.9 \times 10^{-12}]$ $\text{p}K_a = -\log[7.9 \times 10^{-12}] = 11.1$	determines the pKa is 11.1 [1 mark]	
3b)	When $[\text{H}^+]$ is 10 times greater than the K_a value of alizarin yellow R the In^- form dominates and the indicator is yellow. When $[\text{H}^+]$ is 10 times less than the K_a value of alizarin yellow R the HIn form dominates, and the indicator is red. As indicators change colour at the pH at which the $[\text{HIn}] = [\text{In}^-]$, the pH range of the colour change for alizarin yellow R is its $\text{p}K_a \pm 1$.	- explains when $[\text{HIn}]:[\text{In}^-] > 10$ the solution is yellow and when $[\text{In}^-]:[\text{HIn}] > 10$ the solution is red [1 mark] - explains that indicators change colour when $[\text{HIn}] = [\text{In}^-]$ [1 mark] - identifies that the pH range of colour change is $\text{p}K_a \pm 1$ [1 mark]	For <i>explains that indicators change colour when $[\text{HIn}] = [\text{In}^-]$</i> also accept when pKa equals pH.
3c)	Alizarin yellow R would be a suitable indicator for a weak acid–strong base titration.	identifies weak acid–strong base titration [1 mark]	Allow FT mark for correct titration based on incorrect pH range.

Q	Sample response	The response:	Notes
4a)	$\text{HOCl(aq)} \rightleftharpoons \text{H}^+(\text{aq}) + \text{OCl}^-(\text{aq})$ $\text{HCl(aq)} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	- describes that HOCl partially dissociates using reversible arrow [1 mark] - describes that HCl fully dissociates using forward arrow [1 mark]	Also accept equation without states.
4b)	$K_a = \frac{[\text{H}^+][\text{OCl}^-]}{[\text{HOCl}]}$ $x = [\text{OCl}^-] = [\text{H}^+]$ $2.8 \times 10^{-8} = \frac{x^2}{2.50}$	identifies that $[\text{H}^+] = [\text{OCl}^-]$ [1 mark]	
	$x = [\text{H}^+] = 2.6 \times 10^{-4}$	determines that $[\text{H}^+]$ is 2.6×10^{-4} [1 mark]	$[\text{H}^+] = 2.6 \times 10^{-4}$ is implied by $\text{pH} = 3.6$.
	$\text{pH} = -\log[\text{H}^+] = 3.6$	calculates pH [1 mark]	Allow FT mark for pH correctly calculated from incorrect $[\text{H}^+]$.

Q	Sample response	The response:	Notes
5a)	Spot P is glutamic acid (Glu), Q is cysteine (Cys) and R is arginine (Arg). At pH 5, Glu is negatively charged and so would move towards the positively charged electrode. This is because the buffer's pH is higher than the pH of Glu's isoelectric point. At pH 5, Arg is positively charged and so would move to the negative electrode. This is because the buffer's pH is much less than the pH of Arg's isoelectric point. Cys has little charge due to the pH of the buffer and isoelectric point being similar. This means that the molecule would not move very far towards the negative electrode.	- determines that spot P is Glu, spot Q is Cys and spot R is Arg [1 mark] - explains link between the buffer's pH and the pH of the isoelectric point of Arg and Glu [1 mark] - explains link between the charge of Arg or Glu and their positions in the gel [1 mark] - explains link between the buffer's pH and the pH of the isoelectric point of Cys and its position in the gel [1 mark]	Allow FT mark/s for explanations of positions correctly based on incorrect charges. For <i>Cys's position in the gel</i>, also accept: Cys would be mostly zwitterions, so each end would be attracted to the oppositely charged end of the gel. Therefore, minimal movement would occur.
5b)	Spot X is between spot Q and R, so it must be from an amino acid with an isoelectric point between 5 and 10.7. Spot X is closer to spot R; therefore its isoelectric point is close to 10.7. Therefore, spot X is lysine, with a pH of 9.7 at its isoelectric point.	- identifies spot X as Lys [1 mark] - explains link between the pH of the isoelectric point of the amino acid chosen with reference to the isoelectric point of Cys or Glu or Arg [1 mark]	Allow FT mark for correct identification of X based on incorrect identification from 5a).

Q	Sample response	The response:	Notes
6a)	Hexane has a higher boiling point than pentane because a longer carbon chain has a larger molar mass, which increases dispersion forces.	• predicts that hexane has a higher boiling point [1 mark] • explains that increasing chain length increases dispersion forces [1 mark]	For <i>increasing chain length</i>, also accept increasing molar mass. For <i>dispersion forces</i>, – accept: ▪ London forces ▪ van der Waals forces ▪ temporary dipole–dipole forces. – do not accept: ▪ dipole–dipole intermolecular forces.
6b)	Butanal is polar, and pentane is non-polar. Butanal forms dipole–dipole interactions that are stronger than the dispersion forces in pentane; therefore, butanal has a higher boiling point.	• identifies that butanal is polar and/or has dipole-dipole forces [1 mark] • identifies that pentane is non-polar and/or only has dispersion forces [1 mark] • explains that butanal has a higher melting point due to stronger intermolecular forces [1 mark]	
6c)	Propanoic acid and butanol are highly soluble in water because the polar COOH and OH groups can form hydrogen bonds with water.	• identifies that propanoic acid and butanol are highly soluble in water [1 mark] • explains that solubility is due to hydrogen bonding with water [1 mark]	For <i>propanoic acid</i>, also accept: – CH₃CH₂COOH – carboxylic acid. For <i>butanol</i>, also accept: – CH₃CH₂CH₂CH₂OH – alcohol.
6d)	Primary alcohol. The OH group is attached to a carbon that is attached to one carbon atom.	• identifies primary alcohol [1 mark] • explains that the OH group is attached to a carbon that is attached to one carbon atom [1 mark]	For <i>attached to one carbon atom</i> , also accept attached to a terminal carbon atom.
6e)	ethyl methanoate	• names ester as ethyl methanoate [1 mark]	

Q	Sample response	The response:	Notes											
7	<table><tr><th rowspan="2">Contrast</th><th colspan="2">Hydrogen fuel cell</th></tr><tr><th>Acidic conditions</th><th>Alkaline conditions</th></tr><tr><td>Electrolyte used</td><td>H⁺</td><td>KOH</td></tr><tr><td>Movement of ions</td><td>H⁺ ions move through the proton membrane towards the cathode</td><td>OH⁻ ions move through the membrane towards the anode</td></tr></table>	Contrast	Hydrogen fuel cell		Acidic conditions	Alkaline conditions	Electrolyte used	H ⁺	KOH	Movement of ions	H ⁺ ions move through the proton membrane towards the cathode	OH ⁻ ions move through the membrane towards the anode	- identifies that an alkaline fuel cell has an alkaline (KOH) electrolyte [1 mark] - identifies that an acidic fuel cell has an acidic (H⁺) electrolyte [1 mark] - identifies that OH⁻ ions move through the membrane towards the anode [1 mark] - identifies that H⁺ ions move through the proton membrane towards the cathode [1 mark]	For <i>acidic electrolyte</i>, also accept a proton exchange (H⁺) membrane. For <i>OH⁻ ions move through the membrane</i>, also accept: – oxidation half-equation $2\text{H}_2(\text{g}) + 4\text{OH}^-(\text{aq}) \rightarrow 4\text{H}_2\text{O}(\text{l}) + 4\text{e}^-$ – reduction half-equation $2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) + 4\text{e}^- \rightarrow 4\text{OH}^-(\text{aq})$ For <i>H⁺ ions move through the proton membrane</i>, also accept: – oxidation half-equation $2\text{H}_2(\text{g}) \rightarrow 4\text{H}^+(\text{aq}) + 4\text{e}^-$ – reduction half-equation $4\text{H}^+(\text{aq}) + 4\text{e}^- + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{l})$
Contrast	Hydrogen fuel cell													
	Acidic conditions	Alkaline conditions												
Electrolyte used	H ⁺	KOH												
Movement of ions	H ⁺ ions move through the proton membrane towards the cathode	OH ⁻ ions move through the membrane towards the anode												

Q	Sample response	The response:	Notes
8a)	Compound A Structural formula <pre> H H H H — C — C — C — C ≡ N H H H </pre> IUPAC name: butanenitrile Compound D Structural formula <pre> H H H H — C — C — C — OH H H H </pre> IUPAC name: 1-propanol	- determines structural formula for compound A [1 mark] - determines that the IUPAC name for compound A is butanenitrile [1 mark] - determines structural formula for compound D [1 mark] - determines that the IUPAC name for compound D is propanol [1 mark]	For <i>structural formula for compound A</i>, also accept CH₃CH₂CH₂CN. Allow FT mark/s for correct IUPAC names based on incorrect structural formulas. For <i>structural formula for compound D</i>, also accept CH₃CH₂CH₂OH. For <i>IUPAC name for compound D</i>, also accept: – propan-1-ol – propanol.
8b)	propene	determines that the IUPAC name for the alkene is propene [1 mark]	For <i>IUPAC name</i>, also accept: – prop-1-ene – 1-propene.
8c)	addition reaction	identifies addition reaction [1 mark]	
8d)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + 2\text{H}_2 \xrightarrow[\text{Ni}]{\text{heat}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$	- provides a balanced chemical equation for converting butanenitrile to butanamine [1 mark] - identifies an appropriate catalyst [1 mark]	Also accept equation with states. Allow FT mark for correct balanced chemical equation that demonstrate conversion of a nitrile to an amine based on incorrect response to 8a). For <i>catalyst</i>, also accept: – Pd – LiAlH₄.
8e)	Amine B: butanamine; Amine C: propanamine. Amine B has one extra carbon in its carbon chain.	identifies that amine B has an extra carbon in its chain [1 mark]	

Q	Sample response	The response:	Notes
9a)	K_c is greater than 1; therefore, the equilibrium lies towards the product, so the solution will be orange.	- explains K_c is greater than 1; therefore, equilibrium lies towards the product [1 mark] - deduces that the solution will be orange [1 mark]	Allow FT mark for correct deduction based on incorrect identification that K_c is less than 1.
9b)	Let $[H^+] = x$ $K_c = \frac{[Cr_2O_7^{2-}]}{[CrO_4^{2-}]^2 [H^+]^2}$ $4.2 \times 10^{-14} = \frac{6.0 \times 10^{-2}}{(1.2 \times 10^{-3})^2 (x)^2}$	provides appropriate substitution [1 mark]	
	$x^2 = 1.0 \times 10^{-10}$ $x = 1.0 \times 10^{-5}$	determines $[H^+]$ [1 mark]	Allow FT mark for correct $[H^+]$ based on incorrect substitution.
	$pH = -\log(10^{-5}) = 5$	calculates pH is 5 [1 mark]	Allow FT mark for correct pH based on incorrect $[H^+]$.



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